SAM AUTOMATION

Business Understanding Milestone Presentation

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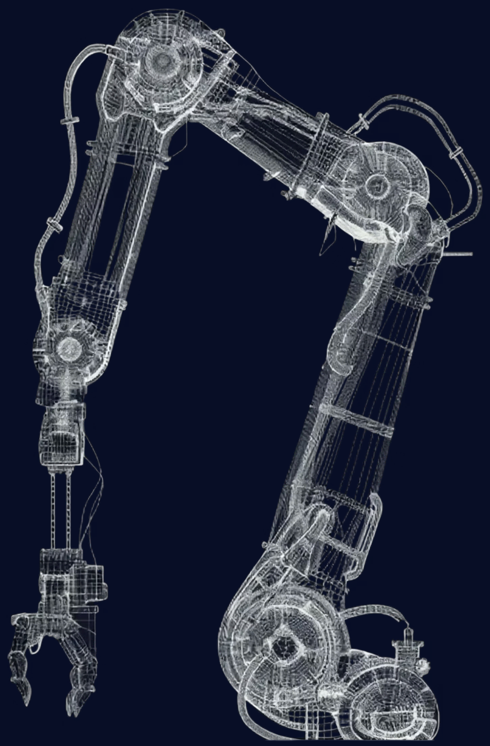
Content



Executive summary

Brief overview

- Current situation
- Proposed approach:
  - Monitor the health of the robots in real-time
  - Reduce potential system failure before they happen
  - Schedule maintenance proactively

Key value proposition for stakeholders

- Enhanced reliability
- Cost reduction
- Improved client trust
- Competitive advantage

Approval

- Collect necessary data
- Allocate resources
- Initiate development

Problem understanding & research insights

Key Research Insights

- **Significant economic impact**
  - 3x more repair costs
  - Losing 3-5% of production capacity

(Bocewicz et al., 2024)

- **Measurable results**
  - 50% less downtime
  - 10-40% cost savings

(McKinsey & Company, 2021)

- **Mature & accessible technology**
  - 70min early detection
  - Edge computing ready

(Givnan et al., 2021)

Evidence Supporting Project Need

- **Changing market**
  - Competitors advancing
  - Client expectations rising
- **The technology has been proven to work**
  - Works with normal data
  - Edge deployment ready
- **The benefits**
  - Less downtime
  - Stronger client trust
  - Jan 2026 achievable

Refined Problem Statement

Unused data without predictive capabilities

Proposed approach & methodology

1. **Overall Approach**

   Raw robot data → Actionable maintenance insights

2. **Methodology**

   Robot & data collection → Data processing & modeling → Dashboard & alerts

3. **Timeline**

   Business understanding → Lab setup → Data understanding → Model development & data engineering → Tool refinement & testing → Prototype → Finalization & scale-up

4. **Goal**

   Functional demo, datasets, pipeline, roadmap for scaling

Business Requirements Document overview

Purpose of the BRD

Defines project boundaries, Live document, Locks expectations & prevents scope creep

Use in Project Execution

Everything traces back to BRD, Phase 2 List, Out of Scope

Key Requirements

Data collection, Anomaly detection model, Inference pipeline, User-friendly dashboard

Success Criteria

Two distinct fault types detectable, ≥70% of alerts lead to action in the pilot, slow-end response time is ≤ 500 ms during runs

Resource requirements & risk management

Resource Requirements

People:

- Expert availability (weekly)

Access:

- Robot
- Cloud
- Sensors

Risk Management

- Robot availability delays
- Data differing from client environment
- Tool variability across devices
- Scope creep
- Firmware updates

Next steps & approval request

Immediate next steps

- Lab setup + robot instrumentation
- Data collection on pressure, vibration, temperature + simulated faults
- Data Quality Report + confirmed signal set

Weeks 8–10

- Build ingestion pipelines, feature engineering
- Train/evaluate anomaly detection models
- Model cards + pipelines v1

Weeks 11–14

- Technician-facing dashboard with live data
- Pilot tests on second robot
- Prototype v1.0 presentation + feedback

Weeks 15–18

- Finalization, scale-up roadmap, and system packaging
- Prototype v2.0 + launch presentation

Reference List

Bocewicz, G., Frederiksen, R. D., Nielsen, P., & Banaszak, Z. (2024). Integrated preventive-proactive-reactive offshore wind farms maintenance planning. Annals of Operations Research. https://doi.org/10.1007/s10479-024-05951-4

Givnan, S., Chalmers, C., Fergus, P., Ortega, S., & Whalley, T. (2021, October 1). Real-Time Predictive Maintenance using Autoencoder Reconstruction and Anomaly Detection. ArXiv.org. https://doi.org/10.48550/arXiv.2110.01447

McKinsey & Company. (2021, July 22). Prediction at scale: How industry can get more value out of maintenance | McKinsey. Www.mckinsey.com. https://www.mckinsey.com/capabilities/operations/our-insights/prediction-at-scale-how-industry-can-get-more-value-out-of-maintenance

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